Open-Source Procedural Radio Propagation Modelling for Sensor Networks in Smart Agriculture

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***Abstract*—Wireless sensing for smart agriculture, such as in greenhouses and vertical farms, must operate through crop canopies whose geometry and electromagnetic properties change with growth and water content. The same propagation conditions affect sensor connectivity, radio frequency (RF) energy delivery, and the feasibility of sensing the crop through changes in the wireless channel. This paper presents HOUSIONNA, a workflow connecting procedural scene generation in Houdini with GPU-accelerated radio ray tracing in Sionna RT. Batch execution supports systematic exploration of alternative crop and infrastructure configurations. The evaluation stage combines visual inspection of paths and radio maps with application metrics for sensor coverage and RF energy harvesting. The workflow provides a reproducible simulation foundation towards agricultural digital twins.**

***Index Terms*—Smart agriculture, environmental monitoring, Internet of Things, sensor networks, RF energy harvesting, flexible plant sensors, wireless sensing, digital twins, ray tracing.**

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Fig. 1: Example of a procedurally generated cultivation scene in Houdini, with plants on trays, infrastructure, and devices.

I. Introduction

Controlled Environment Agriculture (CEA) relies on sensing and control to manage crop production in greenhouses and vertical farms. Spatial microclimate differences motivate measurements at multiple locations [1], while plant-mounted sensors provide observations of the crop itself. Leaf movement measurements can support studies of abiotic stress [2], and flexible sensors can monitor plant conditions [3]. An RF-powered batteryless plant movement sensor combines inertial measurements with Bluetooth Low Energy (BLE) telemetry [4]. Such devices must accommodate plant movement and growth, favouring small nodes with limited energy storage. Their placement is driven by the measurement task, which may put them deep inside a canopy. We use sensor node for a measurement device and gateway for the network infrastructure, regardless of link direction.

These applications impose different RF requirements. Active telemetry needs sufficient link margin, and batteryless operation also requires enough harvested energy per observation. Backscatter reduces the need for a sensor node to generate its own carrier [5], but requires both illumination and a return link to the gateway. Distributed wireless power transfer (WPT) depends on spatial channels and coherent combining [6]. The propagation channel can also provide observations of the environment, as illustrated by radio tomography [7]; relating channel changes to crop conditions requires crop-specific models and independent reference measurements.

The cultivation environment affects all these links. Leaves, stems, growing trays, support structures, and greenhouse surfaces alter line-of-sight visibility and multipath. Growth changes the geometry, while water content changes the electromagnetic properties of plant tissue. Measurements in a tomato greenhouse show substantial differences from conventional propagation models [8]. Deployment analysis therefore needs site-specific channels and should account for changes in the crop, device positions, and infrastructure.

Procedural scene generation makes these comparisons practical. An example is shown in Fig. 1. Parameterised models can vary plant spacing, canopy dimensions, leaf orientations, growth stage, and device placement without manually rebuilding each scene. Explicit parameters make variations repeatable and enable controlled sweeps in which selected properties change while the remaining assumptions stay fixed. The same scene family can then support communication and energy analyses, exposing placements that work for one crop configuration but fail as the canopy develops. This reduces scene-preparation

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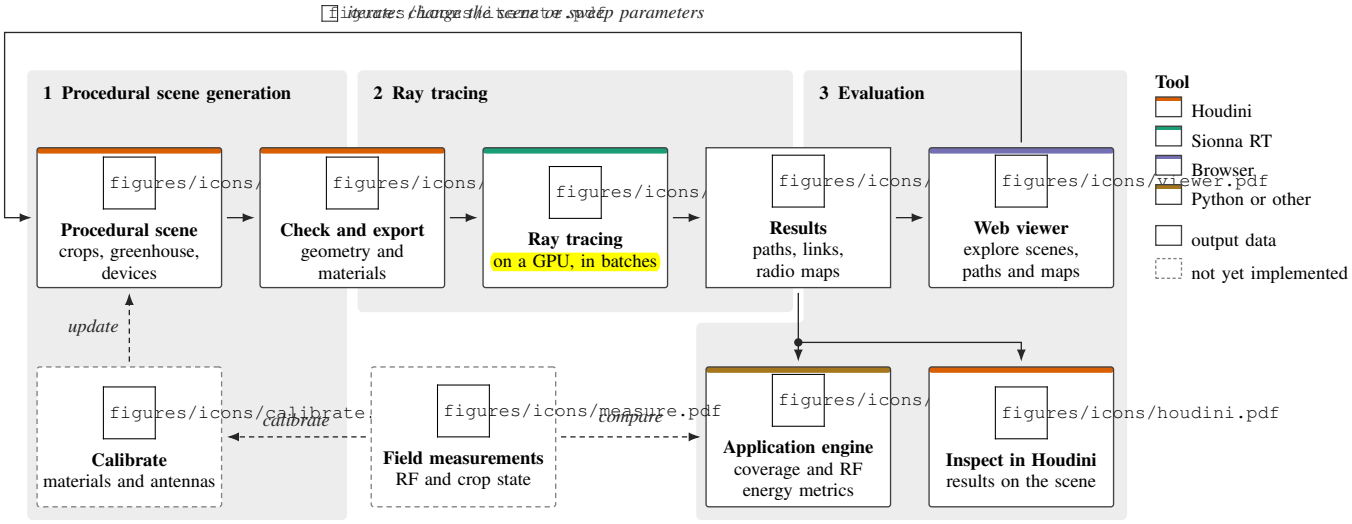


Fig. 2: The HOUSIONNA workflow in three stages. *Check and export* and *Results* hand over between stages; the bar on each step shows the tool that performs it, and the square box is output data. Dashed elements are needed for an agricultural digital twin, in which field measurements calibrate the materials and antennas of the scene. Each box and stage title links to the section that describes it.

effort and supports reproducible comparisons of sensor node and gateway layouts.

This paper presents HOUSIONNA, a workflow connecting procedural scene generation in Houdini with GPU-accelerated radio ray tracing in Sionna RT [9]. It integrates scene preparation and export, batch execution, run provenance, visual inspection, and application analysis; Sionna RT provides the electromagnetic solver [10], [11]. Paths, link metrics, and radio maps can be inspected in Houdini or a web viewer and used to evaluate sensor coverage and RF energy harvesting. A batteryless plant-sensor use case brings these stages together to examine connectivity, charge time, and the choice of harvester

Figure 2 outlines the workflow. Sections II to IV describe procedural scene generation, ray tracing, and evaluation, respectively. Section V applies the workflow to the plant-sensor use case, and Section VI discusses limitations and the steps towards an agricultural digital twin.

II. Procedural Scene Generation

The first stage of HOUSIONNA constructs the agricultural environment in Houdini. A parameterised node network generates geometry, assigns material attributes, and places gateways, sensor nodes, and measurement surfaces for the ray-tracing stage. It can represent alternative crop layouts or infrastructure configurations without manually rebuilding every scene.

A. Agricultural Geometry

The scene parameters include plant spacing, canopy height and depth, leaf orientations, infrastructure layout, and device positions; gateways and sensor nodes can be placed anywhere, including on plants and on mobile robots. Plants are generated with L-systems in three growth stages, optionally with fruit, so growth is represented by a sequence of static scenes. People, autonomous robots, windows, and metal frames can be added as disturbances of the channel. A geometric growth parameter is not automatically a validated physiological model. Figure 4 illustrates the three growth stages. *gilles* be sure to mention where these models are coming from.

B. Radio Materials

The built-in ITU-R P.2040 materials of Sionna are bounded in frequency [12]: apart from wood and glass, they are defined only from 1 GHz, which excludes the 868 MHz and 915 MHz bands. HOUSIONNA therefore adds frequency-dependent models and records a source and status for each material: Cole–Cole fits of common building materials measured from 0.2 GHz to 67 GHz [13], conductors for metal structures, water and nutrient solution [14], soils [15], and the dual-dispersion vegetation model for leaves and stems [16]. The vegetation coefficients are not yet verified, and fruit and growing media use placeholder values, so plant materials must be treated as provisional. ITU-R P.833 provides an independent vegetation-attenuation reference for appropriate scenarios [17]; it does not replace calibration for the specific crop and geometry.

III. Ray Tracing

The second stage transfers the procedural scene to Sionna RT, computes propagation paths and radio maps, and records each run for subsequent inspection and analysis.

A. Scene Checks and Export

The exporter groups triangular geometry by object name and writes meshes, a scene description, material parameters, device definitions, and solver settings.

Export checks include triangulation, object names, material consistency, and device definitions. These are checks of simulation input integrity; they do not establish biological realism or material accuracy. A preview stage loads the exported scene in Sionna and returns its geometry for inspection in Houdini.

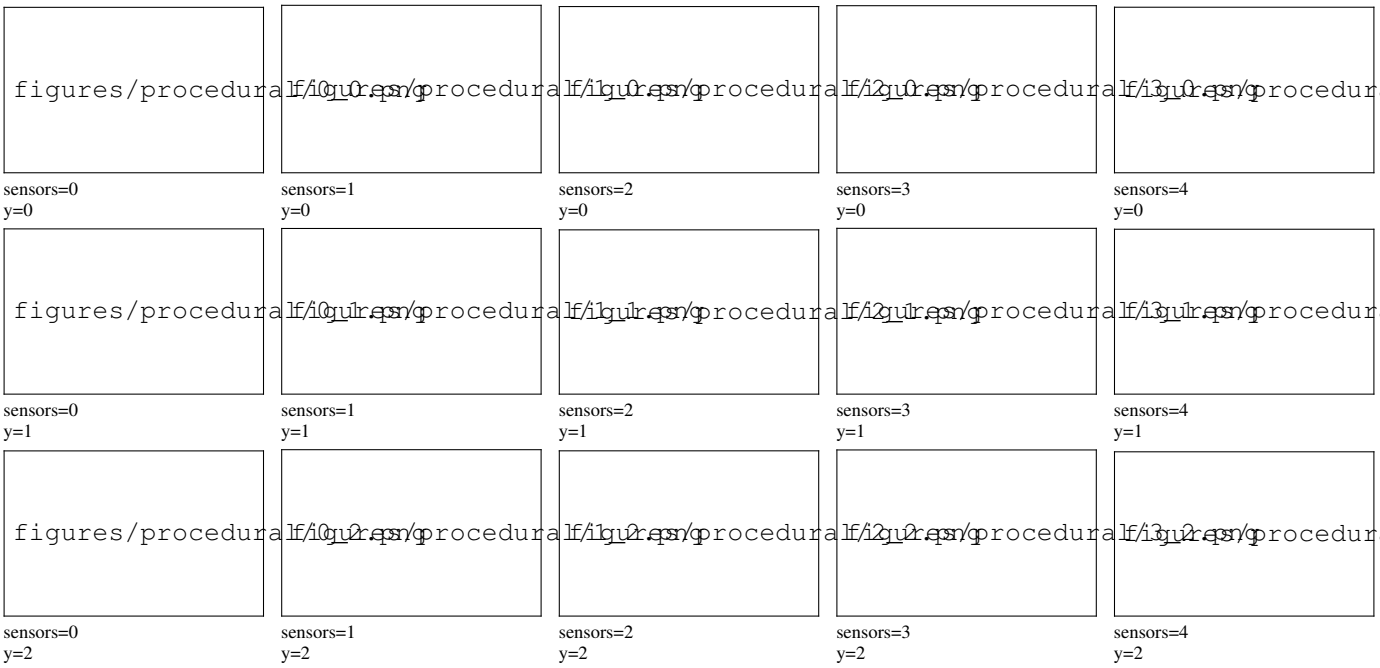


Fig. 3: Procedural scene variations.



Fig. 4: Three generated growth stages of strawberry (*Fragaria* × *ananassa*, top) and tomato (*Solanum lycopersicum*, bottom).

This helps identify coordinate, scale, and export discrepancies before interpreting propagation results.

B. Solver Configuration and Geometry Interactions

The current implementation uses Sionna RT 2.2.0 and a single isotropic, vertically polarised antenna at each gateway and sensor node [9]. The solver accepts controls for line of sight, specular and diffuse reflection, transmission/refraction, diffraction, interaction depth, and sampling budgets. Enabling an interaction does not guarantee that a chosen scene representation models the corresponding canopy physics accurately.

Sionna models interactions with surfaces: each surface a ray crosses acts as a slab with the full thickness of its radio material. A closed canopy shell, internal sheets, and explicit leaf geometry can produce different depth-dependent attenuation. Refining the mesh alone does not resolve this modelling choice. When a leaf or stem is small relative to wavelength, a surface ray approximation may miss relevant scattering behaviour. Sionna RT also discards paths containing a segment shorter than $1\,\mathrm{cm}$, so every path through a closed leaf or stem thinner than this is lost. The slab behaviour and the $1\,\mathrm{cm}$ limit were confirmed with analytical reference cases (free space, a single slab, a closed block, and thin glass) included in the repository [9]. Representation and material uncertainty must be assessed together.

C. Execution and Reproducibility

The solver runs in a separate Python environment, locally on an NVIDIA GPU or a CPU under Linux, or on a remote GPU host. Results are written as compressed NumPy arrays for the evaluation stage. This separation allows the modelling workstation and compute machine to differ.

For parameter sweeps, Houdini task-operator (TOP) work items, one per parameter combination, are exported to individual run folders. A background batch process solves work items sequentially within one Sionna session. Successful items can be skipped when their recorded solver settings match, and failed items are recorded for later retry.

Run records include the exported scene, resolved material properties, devices, settings, a solver copy, and software provenance. Batch records also preserve work-item parameters.

Recording these assets makes the simulated geometry inspectable even after the procedural node network has changed. The records do not yet include the random seed of the path solver or hardware details, which a complete release should add.

IV. Evaluation

The third stage makes the simulated channels available for inspection and application analysis. Results can be examined in Houdini or in a web viewer, while an application engine converts the propagation outputs into relevant application metrics, e.g., coverage and energy metrics.

A. Returned Quantities

Propagation paths are imported as polylines carrying gateway and sensor node identifiers, complex amplitudes, delays, interaction types, and object associations. Per-link outputs include path counts, coherent and incoherent gains, RMS delay spread, and frequency-response magnitudes. Radio maps provide path gain per gateway over triangular measurement surfaces. Radio maps average power over cells and do not retain phase, but under maximum-ratio transmission the local mean RF power at the sensor node is $P_{\mathrm{GW}} \sum_i g_{ij}$, where P_{GW} is the total RF power budget of the gateways. Mean RF power can therefore be evaluated over an area from the per-gateway maps. Maximum-ratio transmission assumes ideal channel knowledge at all gateways and is an upper bound for distributed operation [6].

B. Visual Inspection and Run Browsing

The compressed result arrays are imported into Houdini as attributed geometry, allowing paths and radio maps to be inspected in the original scene. A web viewer built on three.js displays the same paths, radio maps, and link metrics without a Houdini installation, steps through the runs of a batch, and can be published as a read-only website. This allows results to be disseminated more conveniently without relying on Houdini.

C. Application Engine and Metrics

The application engine converts the raw RF propagation channels between gateways and sensor nodes into metrics relevant for different applications and technologies. For instance, it can evaluate the energy harvesting performance of the system as the plants grow.

1) Radio Coverage: Coverage can be evaluated at specified sensor locations or over a measurement surface using Sionna RT's radio map solver. Both approaches can be repeated across crop configurations to assess how coverage changes as the plants grow, but they answer different placement questions.

Per-link evaluation: explicit sensor nodes are placed at candidate sensing locations, including on leaves or within the canopy. The path solver returns the paths between each gateway and each sensor node, with the configured antenna patterns and orientations. For crop configuration s, the incoherent power gain from gateway i to sensor node j is

$$g_{ij}^{(s)} = \sum_k \left| a_{ijk}^{(s)} \right|^2, \tag{1}$$

where $a_{ijk}^{(s)}$ is the coefficient of path k. The effective path loss to the gateway with the largest gain is $L_{\mathrm{best},j}^{(s)} = -10 \log_{10}\left(\max_i g_{ij}^{(s)}\right)$ in decibels. Summing path powers omits their phase cross terms, so this metric excludes fast fading. The complex path coefficients and delays can instead be used to reconstruct the coherent channel and examine constructive and destructive interference at a particular sensor position. Per-link evaluation therefore supports detailed checks of actual mounting positions and antenna orientations, but assesses only the sensor node locations supplied to the solver.

Radio maps: the solver estimates the average non-coherent path gain from each gateway over cells of a measurement surface [11]. Sionna supports rectangular cells on a plane or triangular cells on a mesh; HOUSIONNA uses the latter. This provides an overview of candidate deployment regions without placing an explicit sensor node at every location. Coarse cells can conceal local shadowing, whereas finer cells resolve smaller features but require more ray samples for stable estimates. [1]

For either approach, coverage requires a device-specific link budget. Using the selected point gain or cell-average gain, the link margin is

$$M_{ij}^{(s)} = P_{\mathrm{GW},i} + 10 \log_{10} g_{ij}^{(s)} - S_j - L_{\mathrm{impl}} - M_{\mathrm{reserve}}, \tag{2}$$

where $P_{\mathrm{GW},i}$ and sensor node sensitivity S_j are in dBm, and L_{impl} and M_{reserve} are implementation losses and a design reserve in decibels. A non-negative margin indicates coverage under these assumptions. When gateway powers differ, the best gateway maximises the margin rather than the path gain alone. For an uplink, reciprocity allows the same point-link gain to be used with the sensor node's RF power and the gateway's sensitivity, provided the antenna configurations are unchanged.

Across a set of crop configurations, the worst margin $\min_s \min_j \max_i M_{ij}^{(s)}$ tests whether every sensor location has an adequate gateway in every simulated configuration.

2) Energy Harvesting over RF: For WPT, RF power at a sensor node must be combined with a measured rectifier or harvester response. Practical responses are nonlinear, with an operating threshold, power-dependent conversion efficiency, and saturation at high input powers [19]. Measurements [20] can parameterise $F_{\mathrm{EH}}(P_{\mathrm{RF}}, f)$, which maps RF input power and frequency to usable DC power under specified load conditions.

During charging, the changing buffer voltage can also affect the conversion response [21]. For a given waveform and harvester, we therefore include the buffer voltage $V_{\mathrm{buf}}(t)$ in the response and write the charging energy balance as

$$\int_0^T F_{\mathrm{EH}}(P_{\mathrm{RF}}(t), f, V_{\mathrm{buf}}(t))\, dt \geq \Delta E_{\mathrm{buf}} + E_{\mathrm{loss}}(T), \tag{3}$$

[1] A cell average does not establish that every location within that cell is covered, and reducing the cell size does not restore multipath phase interference. Moreover, Sionna RT's radio maps do not apply the configured sensor node antenna pattern or combining vector [18]. Their gains must therefore be interpreted with care when assessing a plant-mounted antenna with a particular orientation or polarisation.

where ΔE_{buf} is the required increase in stored energy from the initial state to the target operating level, sufficient for a measurement-and-transmission cycle. $E_{\mathrm{loss}}(T)$ accounts for leakage and quiescent consumption not already included in F_{EH}.

The charge time is the earliest T satisfying this balance and the harvester's startup requirements. Because the response is nonlinear, evaluating it at the average RF input power generally gives a different result from averaging the harvested power. Charging estimates should therefore follow the RF power and buffer state over time. For a static channel and a response measured at a single output voltage, as for datasheet curves, the balance reduces to a constant-power charge time $T = \Delta E_{\mathrm{buf}}/F_{\mathrm{EH}}(P_{\mathrm{RF}}, f)$, an approximation used in Section V.

V. Use Case

The use case evaluates a batteryless plant sensor such as that of [4] in a simulated cultivation room. The sensor harvests energy at 915 MHz and reports over BLE at 2.4 GHz, so the same scenes are traced at both frequencies, and at 5.18 GHz for comparison. The metrics follow Section IV-C.

Crop and Cultivation Environment. The scene is a 5.8 m by 5.8 m cultivation room with trays on legs, a structural frame, sprayers, windows, a robot, and people. Three plant generations are evaluated, with canopy heights of 0.17 m, 0.37 m and 0.68 m. The plant materials use the provisional vegetation model of Section II-B. *gilles* add picture of a snapshot of the scenario, can be removed later if needed. or reused in previous sections.

Sensor Node and Gateway Locations. Each plant generation is a bed of plants, with nine sensor nodes placed on the plants at positions generated with the scene, between 5 cm and 29 cm above the growing tray. A single gateway, 2 m above the trays near the centre of the room, powers the sensor nodes and collects their reports. Every sensor node was checked for a surface within 1 cm in the direction of the gateway, where Sionna RT discards the path segments (Section III); this applies to two of the 27 sensor nodes, both of generation 1, whose gain dropped by up to 40 dB, so they are excluded.

Simulation and Device Settings. Each scene is traced at 915 MHz, 2.437 GHz, and 5.18 GHz with isotropic, vertically polarised antennas, line of sight, specular reflection, and refraction, up to five interactions per path, and 10^6 samples per gateway. The gateway provides 30 dBm at 915 MHz. The BLE uplink uses 0 dBm at the sensor node, a sensitivity of -95 dBm at the gateway, and a 10 dB reserve. Two commercial harvesters are compared at the same RF powers (Fig. 5): the e-peas AEM40940 at 921 MHz, designed for low input powers, and the Powercast P1110B at 915 MHz, designed for higher ones. Their DC output is modelled by quadratic fits to the datasheet curves from [20]. No power is harvested below the lowest datasheet point, at -20 dBm for the AEM40940 and -5 dBm for the P1110B, or where a fit is negative. As the datasheet curves are given at a single output voltage, the time to charge a 346 µF buffer from 0.05 V to 4.5 V is approximated at constant power as $t = C\,(V_1^2 - V_0^2)/(2P_{\mathrm{DC}})$, neglecting leakage.

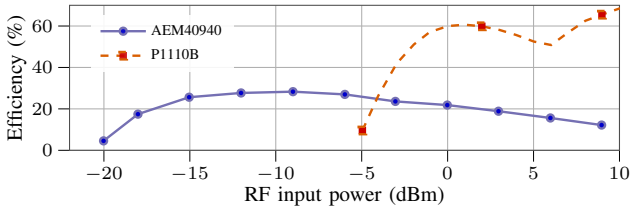


Fig. 5: RF-to-DC efficiencies from the exported harvester measurements [20]. AEM40940 is measured at 921 MHz. The P1110B data is extracted from the datasheets at 915 MHz.

TABLE I: Results per plant generation: median and worst path gain and median RF power at 915 MHz, worst BLE margin at 2.437 GHz beyond the 10 dB reserve, and the number of the n sensor nodes that whose harvester charges the buffer within one hour. Two sensor nodes of generation 1 are excluded because they lie within 1 cm of a leaf.

		Gain (dB)		BLE	P_{RF}	Within 1 h	
Gen.	n	Median	Worst	(dB)	(dBm)	AEM40940	P1110B
1	7	−35.1	−35.6	41.4	−11.5	7	0
2	9	−35.4	−37.7	40.5	−8.3	9	3
3	9	−35.7	−60.0	22.4	−15.2	6	0
Total	25					22	3

A. Deployment Results

Table I summarises the results. The median gain between the gateway and the sensor nodes is -35.3 dB at 915 MHz and drops by 8.2 dB at 2.437 GHz and by 14.5 dB at 5.18 GHz, close to the 8.5 dB and 15.1 dB of free-space scaling. The crop therefore changes the typical link little, as a leaf attenuates by less than 1 dB at 915 MHz, but the tallest canopy shadows individual sensor nodes by up to 25 dB, and the median RF power drops to -15.2 dBm, so that only 6 of its 9 sensor nodes charge the buffer within one hour.

The BLE uplink is not the limiting link: its worst margin is 22 dB beyond the reserve. Energy is, and it depends on the harvester. The median RF power is -10.5 dBm: 23 of the 25 sensor nodes receive more than the -20 dBm at which the AEM40940 starts, and 22 charge the buffer within one hour with it, with a median charge time of 3.5 min, 1.6 min, and 9.8 min for generations 1 to 3. Only 3 receive the -5 dBm that the P1110B needs, all of plant generation 2. The power at the exact sensor node position lies a median of 4.7 dB below its local mean, and up to 16 dB, so small-scale fading decides whether a node operates. A harvester designed for low input powers therefore matters as much as the propagation, and additional, coherently combined gateways can close the remaining gap.

VI. Discussion and Conclusions

HOUSIONNA connects procedural scene generation, GPU radio ray tracing, batch execution, inspection of paths and radio maps in Houdini and a web viewer, and an application engine for communication and WPT metrics. This provides

a practical basis for studying the RF infrastructure serving agricultural sensors across changing crop configurations. Environmental monitoring, plant health monitoring, batteryless IoT, and remote canopy sensing share this propagation foundation while requiring different application models. One finding holds independently of the scene: surface-based ray tracing loses every path through structures thinner than $1\,\mathrm{cm}$, which affects sensor nodes mounted directly on leaves.

Towards an Agricultural Digital Twin. Procedural simulation supports controlled comparisons and reproducible scenario generation, but a digital twin also requires a maintained relationship with the physical environment. In this setting, that relationship would connect observed crop geometry and operating conditions to calibrated RF materials, antenna models, and validated predictions. Periodic updates could then test whether an existing network remains adequate as the crop develops.

Three limits are particularly relevant. First, the present vegetation materials are provisional, and surface-based canopy representations need evaluation across the intended frequencies, in particular for thin stems and leaves. Second, the isotropic single-antenna configuration does not capture plant-mounted antenna detuning, realistic polarisation effects, or array-based power delivery. Third, a sequence of static scenes does not provide continuous state estimation or validated dynamic sensing.

Sionna RT supports differentiable modelling [10], [11], but the file-based wrapper described here does not provide an end-to-end differentiable Houdini pipeline. Gateway placement can be compared by searching subsets of candidate positions on precomputed channels; gradient-based material fitting or continuous placement would need a deliberate integration with the solver. Protocol performance, backscatter modulation, and RF crop inference likewise remain separate modelling tasks.

The broader HyROPA programme includes optical propagation. Here the RF model is the focus; optical energy harvesting and hybrid links are possible later extensions that should reuse compatible geometry and observations without assuming identical interaction models.

Future Work. The next scientific step is to validate canopy representations and material parameters, then evaluate communication, energy, and sensing metrics against physical observations. This progression supports the paper's direction towards agricultural digital twins without treating procedural visualisation alone as a validated twin.

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